Rahul Guruprakash

Firmware & Systems Software Engineer with 4+ years of experience across SoC validation, platform firmware, and embedded Linux

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Education

North Carolina State University **Raleigh, NC**
Master of Science in Computer Engineering *Expected May 2027 | GPA: 4.0/4.0*

R.V. College of Engineering **Bangalore, India**
Bachelor of Engineering in Electrical and Electronics Engineering *August 2021 | GPA: 8.3/10*

Technical Skills

Programming: C, C++, Python, CUDA, OpenMP, AArch64 assembly, Bash, Perl.
Systems: Linux kernel and device drivers, Zephyr, UEFI/EDK II, ACPI, Device Tree, FF-A, TF-A, Arm FVP, QEMU.
Build, Test & Debug: Yocto/OpenEmbedded, Buildroot, CMake, SCons, GCC, Clang/LLVM, GoogleTest, TRACE32, GDB.

Professional Experience

Arm – Intern, Silicon Engineering Team, Austin, TX **May 2026 – Present**

- Defined the system validation strategy for the Coherent Mesh Network (CMN) and AI accelerator; developed Linux kernel and user-space tests on Arm Fixed Virtual Platforms and ported them to silicon.
- Built DDR/HBM traffic tests across platform firmware, Linux, and ML runtime services to reproduce AI workload behavior.
- Authored Yocto layers and recipes to package platform firmware and generate a reproducible Linux root filesystem for SoC QA.

Arm – Engineer, System Compliance (ATG), Bangalore, India **July 2023 – August 2025**

- Delivered Arm's open-source FF-A Architecture Compliance Suite (ACS) releases for FF-A v1.1 and v1.2, adding self-checking tests for secure-partition messaging, memory sharing, interrupts, and notifications.
- Identified FF-A conformance issues across TF-A, Hafnium, and Xen and contributed implementation fixes validated through ACS testing.
- Implemented MMU/GIC support and an SMMUv3 test engine on Arm Base FVP, expanding bare-metal FF-A coverage for memory permissions, interrupt delivery, and DMA isolation.
- Designed a Linux kernel driver and user-space implementation for FF-A ACS, enabling compliance validation from Linux.

Qualcomm – Engineer, RF Software, Hyderabad, India **October 2021 – July 2023**

- Developed RF-control firmware for the Snapdragon X35 5G Modem-RF System across the modem DSP and a dedicated RF-control processor, supporting wireless transceiver image loading, initialization, and RF front-end programming.
- Built GoogleTest-based unit tests for firmware validation, catching functional and integration bugs during virtual-platform testing, FPGA emulation, and silicon bring-up.
- Added CRC validation to RF-control firmware to detect corrupted DMA payloads received from the modem DSP.
- Implemented a firmware image-layout workaround for an instruction-decoder hardware issue and coordinated compiler support with the toolchain team.

Log9 Materials – Engineer, Embedded Systems, Bangalore, India **January 2021 – September 2021**

- Implemented a GB/T 27930 EV charging protocol stack on an Arm Cortex-M MCU, including SAE J1939/21 message handling and multi-packet transport over CAN.
- Stabilized firmware for production on 100+ electric three-wheelers; built a CAN bootloader, test framework, and validation jig.

Academic Research & Projects

Graduate Research Assistant (Part-time) – North Carolina State University, Raleigh, NC **August 2025 – May 2026**

- Owned the hardware architecture and high-speed PCB layout for a custom NVIDIA Jetson Orin NX carrier board for embedded AI biomass mapping; achieved full first-spin bring-up, validating PCIe Gen4, Gigabit Ethernet, and USB 3.2 interfaces.

Computer Architecture Projects – North Carolina State University **Spring 2026**

- **Advanced Microarchitecture (ECE 721):** Implemented stride prediction/recovery and integrated a 30 KB SVP+VTAGE hybrid in an out-of-order RISC-V simulator; raised harmonic-mean IPC by 1.24% over plain SVP across 15 SPEC CPU SimPoints.
- **GPU Architecture (ECE 786):** Extended Accel-Sim with per-SM and per-kernel L1 data cache profiling and selective bypass; achieved a 16.9% geometric-mean IPC gain (2.5–42.8%) over no bypass across four cache-unfriendly workloads.